The Levels of the Theory:

Coarse, Fine, and the Role of the Algebraic Certificate

novaspivack.com/research   P48 monograph (doi.org/10.5281/zenodo.20560550)

UGP Physics Tutorial Series

doi.org/10.5281/zenodo.20657872

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2026

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UGP Physics Tutorial Series

Prerequisites (read first): *The GTE Polynomial: A Step-by-Step Cheat Sheet* · *How the UWCA Works* · *The $\mathbf{\Phi_{MDL}}$ Field: From Discrete Certificate to Continuum Physics*

Next tutorials: *How the Standard Model Quantum Numbers Emerge* · *Gravity from Description-Length Minimization* · *The Born Rule as a Theorem*

See also: *The Three-Tape CMCA* · *Perfect Self-Containment and MDL*

Claim-Strength Taxonomy

Throughout this tutorial, results are labeled with a confidence grade:

- $\mathbf{A}_{\text{Lean}}$ **(CatAL)** — Machine-certified in Lean 4 with zero `sorry` and zero custom axioms. Independently verifiable by anyone with Lean installed.
- **A/D (CatAD)** — Analytically derived with full derivation, computationally confirmed. Not yet machine-certified.
- **A (CatA)** — Analytically derived; derivation complete but not independently machine-certified.
- **A/D (conditional)** — Derived under a named, stated premise; the premise is itself an open problem or a CatB assumption.
- **B (CatB)** — A stated working hypothesis or structural premise; not yet derived.

1 The Central Puzzle: How Many Layers Does the Theory Have?

The question this tutorial answers

Most physical theories have one layer: the equations of physics describe what the universe does. The GTE framework has three or four, depending on how you count. This tutorial explains what those layers are, why they exist, and what each one does. Knowing the layer structure is the key to reading GTE papers correctly — especially the level labels (Level 0, Level 1, Level 2, Level 3) that appear on results throughout the corpus.

If you have read the *Polynomial Cheat Sheet* [1] and the Φ_{MDL} *Field* tutorial [2], you already know the two most important facts:

- The GTE polynomial $p(L, C, R) = C + R - CR - LCR$ over GF(7) is a 19-bit cellular automaton rule — a discrete algebraic object.
- This object is *not the universe*; it is an **algebraic certificate** for the continuum quantum field Φ_{MDL}, which is what the universe is made of.

That two-level picture (discrete certificate vs. continuum field) captures the most important distinction. But the full structure is richer. The certificate itself comes in multiple layers of increasing concreteness, from the raw polynomial all the way to the running tape. And behind the certificate sits a layer of pure arithmetic — the mathematical substrate — that selects and justifies the certificate but is not itself a physical object.

The framework also has a separate organizational vocabulary: the **ontological ladder**, which names the *roles* objects play (what selects, what certifies, what exists), not just their resolution. Understanding how the "levels" vocabulary and the "ladder" vocabulary relate to each other is the core of this tutorial.

The payoff is practical: every result in the GTE paper series is labeled with its level (*Level 0*, *Level 1*, *Level 2*, *Level 3*, or the coarser *Level 1/Level 2*). Reading those labels correctly tells you how strong a result is, whether it is a discrete algebraic fact (machine-certifiable in Lean 4) or a continuum claim (requiring the Lifting Theorem), and whether the claim depends on any physical interpretation beyond the arithmetic.

2 The Coarse Two-Level Scheme

Most GTE papers use a simple two-level vocabulary. This is the right starting point.

> The fundamental distinction (coarse two-level scheme)
>
> **Level 1 — The Algebraic Certificate (discrete):** The CMCA (Chiral Minkowski Cellular Automaton), its PSC-orbit map f_{MDL}, and the 19-bit polynomial p. Discrete, algebraic, machine-certifiable in Lean 4.
>
> **Level 2 — The Physical Substrate (continuum):** Φ_{MDL} — the $\mathbb{Z}_7$-symmetric Klein-Gordon gauge field on $\mathbb{R}^{3,1}$. Continuous, physically real, the actual universe.

2.1 Level 1: The Blueprint

The Level-1 certificate is the **instruction manual** for what the physical universe must contain. It is a finite, discrete, algebraic object. You can run it on a computer. You can write Lean 4 proofs about it. You can enumerate its states and verify its behavior exhaustively.

What the certificate tells you:

- **Which particle types must exist.** Exactly five $\mathbb{Z}_7$ winding sectors pass the PSC filter: $\{0, 2, 3, 4, 6\}$, corresponding to the SM particle classes (`psc_admissible_sectors`, zero sorry).
- **Why there are exactly three generations.** The $\mathbb{Z}_7^5$ orbit under f_{MDL} has exactly three non-vacuum orbit types, cycling with period 3. This is a consequence of the $\mathbb{Z}_7$ arithmetic, not an external input. Machine-certified: `three_generations_physical` (zero sorry).
- **The electric charge formula.** $Q = w_c/3$, where w_c is the centered representative of the $\mathbb{Z}_7$ winding number. Machine-certified: `gte_charge_is_linear_projection` (zero sorry).
- **Topological structure and conservation laws.** Winding-number conservation at every interaction vertex; color confinement; $\theta_{\mathrm{QCD}} = 0$ by three independent algebraic proofs.
- **Computational universality.** The binary restriction of p is Rule 110, the unique CA rule with a machine-certified proof of Turing universality.

All of these facts live at Level 1: they are algebraic facts about the discrete certificate, provable without any reference to the continuum field.

2.2 Level 2: The Building

Level 2 is Φ_{MDL} — the $\mathbb{Z}_7$-symmetric Klein-Gordon gauge field with internal symmetry group $F_{21} = \mathbb{Z}_7 \rtimes \mathbb{Z}_3$, evolving on $\mathbb{R}^{3,1}$.

This is what the universe *is*. Elementary particles are BPS (Bogomolny-Prasad-Sommerfield) topological kink solitons of this field, classified by their $\mathbb{Z}_7$ winding number. Spacetime, quantum mechanics, and all cosmological observables emerge from its dynamics.

What Level 2 adds beyond Level 1:

- Exact Lorentz invariance (any finite-resolution CA has a preferred frame; the continuum field does not).
- Exact particle masses (derived via the orbit cascade and the Self-Referential Renormalization Group; see the *Masses* tutorial [3]).
- Spacetime geometry (3+1D Minkowski spacetime, General Relativity).
- Full quantum field theory (Hilbert space structure, scattering amplitudes).

2.3 A Table: What Lives at Each Level

Result / claim	**Coarse Level**	**Cert**
Three SM generations exist	Level 1	CatAL
Electric charge formula $Q = w_c/3$	Level 1	CatAL
Strong CP angle $\theta = 0$	Level 1	CatAL
Color confinement	Level 1	CatAL
Quantum numbers of all SM fermions	Level 1	CatAL
Winding-sector admissibility ($\{0, 2, 3, 4, 6\}$)	Level 1	CatAL
Fermion / boson statistics	Level 1	CatAL
Turing universality (Rule 110 link)	Level 1	CatAL
Exact Lorentz invariance	Level 2	CatAL
Particle masses	Level 2	CatAD
3+1 spacetime dimensions	Level 2	CatAL
Einstein equations $G_{\mu\nu} = 8\pi G\, T_{\mu\nu}$	Level 2	CatAD
Newton's constant G_N	Level 2	CatAD
Born rule	Level 2	CatAL
Cosmological constant bracket	Level 2	CatAD

The coarse scheme works well for most purposes. Whenever a paper says "Level 1 certificate" or "Level 2 field," it means the CA/algebraic side or the continuum/physical side.

3 The Fine-Grained Four-Level Scheme

For detailed derivations — especially when tracking exactly which structural features come from where — a finer vocabulary breaks the "Level 1 certificate" into three sub-levels.

The GTE Computational Architecture: One System, Four Levels

The polynomial $p(L, C, R) = C + R - CR - LCR$ defines a single physical system expressed at four levels of description; each level is the same physics at a different scale of resolution.

Level 0 — Raw Polynomial (*Object 0*). The k=7, r=1 CA rule over GF(7), as a mathematical object. Class-3 chaotic on generic $\mathbb{Z}_7$ inputs; Class-4 (Rule 110) on $\{0, 1\}$-inputs. MDL + PSC select it uniquely from 7^{343} candidates (`mdl_ca_rule_coding_closed`, CatAL). No dynamics yet; just the formula.

Level 1 — PSC-Projected Orbit Map (*Object 1*, f_{MDL}). The restriction of p to PSC-admissible inputs. f_{MDL} maps 14 of the 343 neighborhoods non-trivially; all others produce the vacuum. On the isolated 5-cell ring, f_{MDL} runs the SM generation orbit $\text{GEN}_1 \to \text{GEN}_2 \to \text{GEN}_3 \to \text{VAC}$ (CatAL). This is the abstract discrete dynamics — orbit structure, quantum numbers, generation count.

Level 2 — CMCA Tape. f_{MDL} embedded in a 1D spatial tape with a chiral pair (Rule 110 forward / Rule 124 backward) and an inner τ_c first-passage clock gating outer updates asynchronously. This is the CMCA: a running, spatially-extended cellular automaton. Three tapes sharing a global outer clock τ_c^{out} give the 3+1D architecture (Dimensional Protocol Principle, CatAL).

Level 3 — Continuum Field (*Object 2*, Φ_{MDL}). The Algebraic Lifting Theorem (`algebraic_lifting_theorem`, CatAL) maps the CMCA's discrete structure to the $\mathbb{Z}_7$-symmetric Klein-Gordon field Φ_{MDL}. SM particles are BPS kink solitons; all SM parameters, spacetime, QM, and cosmology emerge at this level.

Relation to the coarse two-level scheme: Levels 0–2 collectively constitute the *algebraic certificate* — "Level 1" in the coarse scheme. Level 3 here is the *continuum physical substrate* — "Level 2" in that framework. The four-level description is a finer-grained view; both schemes are valid.

3.1 Plain-English Guide to Each Fine Level

Level 0 — The Raw Polynomial

What it is: Just the formula $p(L, C, R) = C + R - CR - LCR$ over GF(7), as a mathematical expression. No spacetime, no dynamics, no orbit — just the arithmetic rule.

What you can prove here: Algebraic identities; the coefficient structure; the link to Rule 110 (when restricted to $\{0, 1\}$ inputs); MDL minimality within the multilinear polynomial class; the fact that exactly four monomials are needed.

Plain English: Level 0 is the recipe card, before you've cooked anything.

Level 1 — The PSC-Projected Orbit Map (f_{MDL})

What it is: The polynomial restricted to the PSC-admissible neighborhood configurations. f_{MDL} turns the abstract rule into a dynamics on the 5-cell ring that generates the Standard Model generation orbit.

What you can prove here: Three-generation structure; Garden-of-Eden stability of gen_1; winding-sector admissibility; charge quantization; fermion/boson statistics from group order; color confinement as MDL-inadmissibility of free colored states; $\theta_{\text{QCD}} = 0$.

Plain English: Level 1 is the first time the recipe actually tells you something about the finished dish: what ingredients must appear, in what combinations.

Level 2 — The CMCA Tape

What it is: The orbit map embedded in a running, spatially-extended automaton — the Chiral Minkowski Cellular Automaton. The tape has the full chiral architecture (Rule 110 + Rule 124) and the inner τ_c first-passage clock. Three tapes sharing a clock give 3+1D.

What you can prove here: Dimensional Protocol Principle (DPP); 3+1D Minkowski structure; SR time dilation; Lorentz algebra $\mathfrak{so}(1,3)$; holographic area scaling; kink-spectrum existence (via the Lifting Theorem).

Plain English: Level 2 is the kitchen — the full apparatus running. The dish is being cooked, but it is not yet the physical meal.

Level 3 — The Continuum Field (Φ_{MDL})

What it is: The unique continuum Lorentz-invariant quantum field whose algebraic structure is certified by Levels 0–2. Φ_{MDL} is a $\mathbb{Z}_7$-symmetric Klein-Gordon gauge field on $\mathbb{R}^{3,1}$ with exact symmetry group $F_{21} = \mathbb{Z}_7 \rtimes \mathbb{Z}_3$. Particles are its BPS kink solitons.

What you can prove here: Exact Lorentz invariance; particle masses; spacetime geometry; gravity; quantum mechanics; cosmological observables.

Plain English: Level 3 is the finished building. This is what exists.

3.2 A Table: Fine-Level Scheme

Fine Level	Name	Key results	Coarse
0	Raw polynomial p	MDL minimality; polynomial form; Rule 110 link	L1 cert
1	f_{MDL} orbit map	Quantum numbers; 3 generations; color confinement	L1 cert
2	CMCA tape	3+1D; SR; holography; DPP	L1 cert
3	Φ_{MDL} field	Masses; GR; QM; cosmology	L2 substrate

When to use which scheme

Use the coarse (two-level) scheme when:

- You are discussing a result in a paper.
- You need to decide: "is this a machine-certifiable algebraic fact, or does it depend on the continuum field?"
- You are reading claims boxes (orange colorboxes in GTE papers).

Use the fine (four-level) scheme when:

- You need to know whether a result depends on the polynomial's structure alone (Level 0), the orbit dynamics (Level 1), the full spatiotemporal tape (Level 2), or the continuum limit (Level 3).
- You are tracing which step in a derivation introduces the first spacetime structure.
- You are reading P41 (CMCA), P45 (three-tape CMCA), or the Mathematical Substrate chapter of P48 [4].

4 The Ontological Ladder

The levels grade *resolution* of description. The ontological ladder grades *role*. These are two orthogonal ways of organizing the same framework.

The Ontological Ladder (canonical definition from P48 Chapter 4)

The theory's objects occupy three rungs, and the rungs must not be conflated.

Rung 1 — Mathematical Substrate. The UGP/GTE arithmetic: the ridge sieve, the integer triples $(a, b, c; g)$ and map T, the generation orbit, and the UWCA evaluator. Pure number theory and combinatorics, prior to any physical interpretation. It is the domain in which both of MDL's selections operate: the seed selection (the Lepton Seed $(1, 73, 823)$) and the rule selection (the polynomial p). *Not a physical object.* Its role is: **to select**.

Rung 2 — Algebraic Certificate. What those selections produce: the 19-bit polynomial p, its PSC-orbit map f_{MDL}, and the CMCA tape (fine Levels 0–2 of the four-level architecture). Machine-checkable in Lean 4. *Not a physical object.* Its role is: **to certify**.

Rung 3 — Physical Substrate. Φ_{MDL}: the $\mathbb{Z}_7$-symmetric Klein-Gordon gauge field on $\mathbb{R}^{3,1}$. *The only rung that physically exists.* Its role is: **to be**.

4.1 Rung 1: The Mathematical Substrate

The mathematical substrate is the number-theoretic ground in which the framework lives before any physical claim is made.

The **ridge sieve** filters which integer triples are arithmetically coherent. At ridge level $n = 10$ (forced by four independent arithmetic certificates), exactly six triples survive; the Minimum Description Length principle selects the unique lexicographically minimal survivor: the Lepton Seed $(1, 73, 823)$.

Applying the GTE map T twice to the Lepton Seed produces the three N_{eff} values 73, 42, 275 — the electron, muon, and tau informational addresses. This three-generation cascade is pure arithmetic: no physics postulated, no parameters chosen.

The **Universal Windowed Cellular Automaton (UWCA)** is the mathematical substrate's native computer — a finite-local evaluator on which Rule 110 demonstrably runs. Its existence shows that the arithmetic is *self-describing*: the same substrate that produces the generation orbit also supports the computation needed to check it.

The key point: the mathematical substrate does not have a "level" in the four-level scheme. The level vocabulary starts at Level 0 (the polynomial), which is already the *output* of the substrate's MDL selection. The substrate is below the bottom of the level ladder.

4.2 Rung 2: The Certificate

The certificate is what the mathematical substrate's MDL selections produce. It spans fine Levels 0–2: the polynomial (Level 0), the orbit map (Level 1), and the running tape (Level 2).

The certificate is not an approximation to the physical field. It is the *proof system* that certifies what the physical field must contain. Run the certificate on a computer and you are running a mechanically verifiable proof about the structure of matter.

An important subtlety: there is a sense in which the certificate is "closer to" the physical field than the mathematical substrate is — the certificate already contains all the algebraic structure of Standard Model physics — but it is not the physical field. The Lorentz invariance that the physical universe exhibits exactly cannot be achieved by any finite-resolution CA (this is the No-CA-Replica Theorem; see §5.3).

4.3 Rung 3: The Physical Substrate

The physical substrate is Φ_{MDL}. It is the only rung that physically exists: the actual quantum field pervading $\mathbb{R}^{3,1}$, whose topological kink excitations are the elementary particles of the Standard Model.

Φ_{MDL} has *exact* Lorentz invariance, which requires the continuum limit $M \to \infty$. It has exact particle masses, exact scattering amplitudes, exact spacetime geometry.

The ladder is orthogonal to the level vocabulary

The four fine levels (0–3) and the three ontological rungs are two different ways of talking about the same framework.

- **Levels** ask: *how coarsely are we describing the system?* (Raw rule? Orbit? Running tape? Continuum limit?)
- **Rungs** ask: *what role does this object play?* (Selector? Certifier? The thing that exists?)

Levels 0–2 all sit on Rung 2 (certificate). Level 3 sits on Rung 3 (physical substrate). Rung 1 (mathematical substrate) sits below the level vocabulary entirely.

4.4 The Two Key Transitions on the Ladder

Rung 1 → Rung 2: MDL Selection. The mathematical substrate does not automatically produce the certificate. Two selections are required, both governed by the MDL principle:

1. *Seed selection:* MDL selects the Lepton Seed $(1, 73, 823)$ from the ridge sieve survivors (`rsuc_theorem`, `mdl_selects_LeptonSeed`, CatAL).

2. *Rule selection:* MDL, applied to the generation orbit's parity shadow, selects the unique polynomial p (`mdl_ca_rule_coding_closed`, CatAL).

The two selections compose sequentially: the seed selection produces the orbit, and the orbit's parity shadow drives the rule selection. The two arms of the framework (the particle *spectrum*, from the orbit; and the particle *dynamics*, from the polynomial) both trace back to one arithmetic substrate and two MDL applications.

Rung 2 → Rung 3: The Algebraic Lifting Theorem. This transition is covered in detail in §6. The Lifting Theorem is the certified bridge that guarantees: every algebraic fact proved in the certificate holds in the physical field.

5 What the Certificate Does — and Does Not Do

5.1 What the Certificate Certifies

The certificate certifies everything that is a discrete algebraic fact about the universe's structure:

- Which topological excitations are PSC-admissible (i.e., which particle types must exist).
- What quantum numbers they carry ($\mathbb{Z}_7$ winding number, $\mathbb{Z}_3$ color charge, generation index).
- Conservation laws at every interaction vertex.
- Color confinement (free colored quarks are PSC-inadmissible: displaying an isolated colored state costs extra MDL bits).
- The strong CP angle $\theta_{\mathrm{QCD}} = 0$ by three independent algebraic proofs from F_{21} group theory.
- Turing universality (the binary restriction is Rule 110; any computation can be simulated).
- 3+1D Minkowski structure via the Dimensional Protocol Principle.

All of these facts hold at *every* lattice resolution $M \geq 1$. They do not depend on the continuum limit. This is the content of the Algebraic Descent Theorem (§7).

5.2 What the Certificate Cannot Certify

What Level 1 cannot certify (directly)

- **Exact Lorentz invariance.** Any finite-resolution CA has a preferred frame. For a CA with M cells per unit length, the Nyquist residual $\varepsilon_0(M) = \pi^2/(3M^2)$ is strictly positive for all finite M. Exact Lorentz invariance requires $M \to \infty$, which is Φ_{MDL}.
- **Exact particle masses.** Masses come from the Level-3 orbit cascade (energy of the Φ_{MDL} kinks) and the SRRG bridge, not from the CA's discrete rule. The CA certifies *which particles exist*; the continuum field gives their masses.
- **Specific spacetime microstructure at Compton scale.** The Lifting Theorem certifies existence of a particle with given quantum numbers, but does not identify which specific CA glider pattern corresponds to the electron at Compton scale. That identification is an open problem.

These properties require the continuum field Φ_{MDL}.

5.3 The No-CA-Replica Theorem

Why can't the universe simply be the cellular automaton? The answer is Lorentz invariance — and it is a theorem, not merely a philosophical preference.

Theorem: No-CA-Replica (CatAL, zero sorry)

For every finite lattice resolution $M > 0$:

$$\varepsilon_0(M) \;=\; \frac{\pi^2}{3M^2} \;>\; 0.$$

No finite-resolution discrete model can achieve exact Lorentz invariance. The continuum limit $M \to \infty$ is the unique exact Lorentz-invariant model in the $\mathbb{Z}_7$ class, and that limit is Φ_{MDL}.

Lean cert: `no_finite_ca_exact_lorentz_replica` (zero sorry, `CMCAContinuumLimit.lean`, ugp-lean).

Plain-English meaning: Any finite cellular automaton has a *lattice*: a preferred grid of cells. That grid defines a preferred frame — the frame in which the grid is at rest. At high velocities, Lorentz boosts make the lattice structure visible, breaking the symmetry of special relativity. The bigger the lattice cells (smaller M), the bigger the violation. At $M = 7$ (the natural scale for the CA), the error is $\sim 6.7\%$. The *only* CA with zero error is the one with infinitely fine cells — which is the continuum field Φ_{MDL}.

What the theorem does NOT say: It does not say the CA is wrong or useless. It says the CA is the wrong *kind* of object to be the physical substrate. The CA is a certificate — a proof engine. A proof engine does not need to be the physical universe; it needs to be correct about what the physical universe must contain. The CA at $M = 7$ certifies algebraic facts (charge quantization, three generations, $\theta = 0, \ldots$) with *zero* error. The 6.7% error applies only to geometrical/Lorentz-invariance properties, not to the algebraic facts.

The blueprint is not wrong; it is not a building

The no-CA-replica theorem does not mean the CMCA is incorrect. It means the CMCA is the wrong *kind* of object to be the physical substrate. The blueprint of a building is not wrong about what the building contains — it just is not made of steel and glass.

6 The Algebraic Lifting Theorem

Theorem: Algebraic Lifting (CatAL, zero sorry, zero custom axioms)

Let B be a PSC-admissible beable with positive dynamical weight. Then B determines a physical observable at the Compton scale carrying the same $\mathbb{Z}_7$ winding number, $\mathbb{Z}_3$ color charge, and generation index as B. In particular: charge quantization, color structure, the three-generation hierarchy, and S-matrix quantum numbers proved at the beable level hold at the physically observable Compton scale.

Lean cert: `algebraic_lifting_theorem` (`LiftingTheorem.lean`, ugp-lean, zero sorry, zero custom axioms).

6.1 What "Lifting" Means

Lifting is the process of taking a result proved in the discrete certificate (Level 1, the CA) and concluding that the corresponding statement holds in the continuum field (Level 2/3, $\Phi_{\rm MDL}$).

A useful analogy from mathematics: completeness theorems in logic. In formal logic, a theory is *complete* if every statement true in the intended model is provable from the axioms. The Lifting Theorem plays an analogous role: every algebraic fact proved in the CMCA (the discrete model) holds in $\Phi_{\rm MDL}$ (the physical realization). The CA is the model; $\Phi_{\rm MDL}$ is the full theory.

More concretely: suppose you prove in the CA that winding number is conserved at every interaction vertex. The Lifting Theorem guarantees that winding number is also conserved at every interaction vertex in $\Phi_{\rm MDL}$. The proof happens in the CA (where it is a finite computation, machine-checkable); the conclusion holds in the physical universe.

6.2 What Can Be Lifted vs. What Cannot

- **Can be lifted:** any property in the $F_{21} = \mathbb{Z}_7 \rtimes \mathbb{Z}_3$ winding sector — quantum numbers, conservation laws, topological charges, generation structure, admissibility conditions, color charges.
- **Cannot be lifted directly:** geometrical properties (Lorentz invariance, scattering amplitudes at high momentum) that depend on the continuum limit. These are not *forbidden* — they are established by the no-CA-replica theorem and the $\Phi_{\rm MDL}$ Lagrangian, not by the Lifting Theorem.

6.3 A Worked Example: The Three Vacua and Color Charges

The $\Phi_{\rm MDL}$ potential has seven degenerate minima (vacua) at $\Phi_k = 2\pi k/7$ for $k \in \{0, 1, \ldots, 6\}$. The three **ground states** of the CA are $\{0, 1, 5\}$ — the three fixed points of the polynomial under iteration.

At Level 0 (raw polynomial), this is a pure algebraic fact about the fixed points of p. Via the Lifting Theorem, this translates to the three *color charges* of QCD: there are exactly three color states because the CA has exactly three ground states. The algebraic structure at Level 0 lifts to the physical gauge structure at Level 3.

Another example: the electron.

Level 1 (certificate says)	**Level 3 (physics says)**
Beable $[1, 5, 2, 2, 1]$ with winding $W = 4$,	$w = 4$ kink in $\Phi_{\rm MDL}$, $Q = -1$,
Garden-of-Eden (no predecessors), fermionic statistics ($\mathrm{ord}(4) = 3$ in $\mathbb{Z}_7^*$)	$J = 1/2$, $m_e = 0.511\,\mathrm{MeV}$

The charge, statistics, and generation stability all follow from Level 1 arithmetic. The mass and the spin assignment follow from the Lifting Theorem combined with the continuum kink spectrum. Neither the charge nor the mass is a free parameter.

6.4 Why the Lifting Theorem Makes Machine-Certified Physics Possible

Before the Lifting Theorem, there would be a gap: you could prove things about the CA, but those proofs would be about an abstract discrete object, not about the physical universe. The Lifting Theorem closes that gap.

Once the Lifting Theorem is in hand, a Lean 4 proof that certifies some algebraic property of the CA *is* a proof about particle physics. This is why GTE papers can state results like: "The electron's charge is -1 (zero sorry, zero custom axioms)." The proof runs in a finite discrete computer, about a finite discrete object; the Lifting Theorem guarantees the conclusion holds in the physical universe.

7 The Algebraic Descent Theorem

Theorem: Algebraic Descent (CatAL, zero sorry)

Every $F_{21} = \mathbb{Z}_7 \rtimes \mathbb{Z}_3$ conservation law, charge quantization condition, and generation structure certified at the discrete CMCA level holds in any continuum realization at any lattice resolution $M \geq 1$.
Charge quantization, color confinement, asymptotic freedom ($b_0 = 7$), and the strong-CP resolution $\theta_{\mathrm{QCD}} = 0$ are M-independent. Only geometric properties (Lorentz invariance, SR accuracy) require $M \to \infty$.

Lean cert: `algebraic_descent_theorem` (`AlgebraicDescentTheorem.lean`, ugp-lean, zero sorry).

7.1 The Other Direction

The Descent Theorem is the converse of the Lifting Theorem. Where Lifting says "certificate $\Rightarrow$ physics," Descent says "physics $\Rightarrow$ certificate."

More precisely: any physical model that respects the F_{21} winding algebra *must also* enforce all the discrete Level-1 constraints. The continuum field cannot escape what the CA forbids.

This has a striking practical consequence. The Descent Theorem is why all the algebraic structure certified in the CA holds for Φ_{MDL} at *every* scale, not just in the continuum limit. Even at $M = 7$ (one cell per lattice unit), the charge quantization, three-generation structure, and color confinement constraints are exact — no approximation is involved.

7.2 Why the Descent Theorem Matters for Lean Certification

The Descent Theorem explains why Lean 4 can certify physical predictions.

- The certification *happens* at Level 1 (the CA is checkable; it is a finite discrete object).
- The Lifting Theorem says: the certificate implies the physics.
- The Descent Theorem says: the physics must respect the certificate.
- Together: the CA and Φ_{MDL} are algebraically equivalent in the SM winding sector — two descriptions of the same object at different scales, related by a certified bidirectional correspondence.

7.3 A Worked Example: Color Confinement

Consider the statement: "An isolated quark cannot be observed as a free particle."

In the CA (Level 1): A free quark would carry winding $w \in \{2, 6\}$. Displaying such a state requires an extra $\log_2 9 \approx 3.17$ MDL bits to specify its color. MDL minimality forbids this: isolated colored states are PSC-inadmissible. Machine-certified: `psc_forbids_free_colored_quarks` (zero sorry).

By Descent to Φ_{MDL}: The Descent Theorem guarantees that the PSC-inadmissibility established at Level 1 holds at every resolution $M \geq 1$, including the continuum. Φ_{MDL} therefore also forbids free colored quarks: color confinement is an exact algebraic fact of the continuum theory, certified by the CA.

What this means: A 19-bit polynomial over a 7-element finite field certifies — with zero sorry and zero free parameters — that quark confinement holds in the physical universe.

8 Level Framing in Practice: How GTE Papers Use These Labels

8.1 Why Level Labels Matter

Every result in the GTE corpus carries a level label. This is not decoration — it is a precision commitment. A Level 0 algebraic fact (a property of the raw polynomial) is maximally strong: it follows from pure arithmetic and can be verified by exhaustive enumeration if needed. A Level 3 claim (about Φ_{MDL}) may require additional continuum assumptions and the Lifting Theorem.

Knowing the level tells you:

1. **How checkable the claim is.** Level 0–2 claims are in principle machine-verifiable in Lean 4. Level 3 claims require the Lifting Theorem and sometimes additional continuum physics arguments.
2. **What the claim depends on.** A Level 0 claim depends only on the polynomial's arithmetic. A Level 3 claim additionally depends on the continuum field structure and the bridge theorems.
3. **How to cite it honestly.** Writing "CatAL" next to a Level 0/1 result tells the reader it is Lean-certified. Writing "CatAD" next to a Level 3 result tells the reader it is analytically derived and computationally confirmed, but the full machine proof runs through the Lifting Theorem.

8.2 Worked Examples of Level Framing

Level 0 example: ground states are $\{0, 1, 5\}$

Claim: The fixed points of $p(x, x, x)$ over GF(7) are exactly $\{0, 1, 5\}$.
Level: 0 (raw polynomial arithmetic).
Why level 0: This is a statement about the formula $p(x, x, x) = x$ over GF(7). No orbit, no tape, no continuum. Checkable by evaluating seven numbers.
Cert: CatAL (decidable by exhaustive enumeration; zero sorry).

Level 1 example: strong CP angle $\theta_{\mathrm{QCD}} = 0$

Claim: The strong CP angle $\theta_{\mathrm{QCD}} = 0$ exactly.
Level: 1 (group theory of f_{MDL} and F_{21}).
Why level 1: Three independent algebraic proofs from the $F_{21} = \mathbb{Z}_7 \rtimes \mathbb{Z}_3$ group structure establish this — no continuum field theory required. The argument uses the orbit map and its symmetry properties.
Cert: CatAL (three independent machine-certified proofs; zero sorry).

Level 2 example: 3+1D Minkowski structure

Claim: Three CMCA tapes sharing a common outer clock produce 3+1D Minkowski spacetime.
Level: 2 (CMCA tape architecture — the Dimensional Protocol Principle).
Why level 2: The claim is about three running tapes sharing a clock, and whether their tensor product gives the correct spacetime. The tape is the Level-2 object; the DPP is a Level-2 theorem.
Cert: CatAL (`dimensional_protocol_principle_master`, `cmca_tensor_product_gives_31d_minkowski`; zero sorry; ugp-lean).

Level 3 example: Einstein's equations

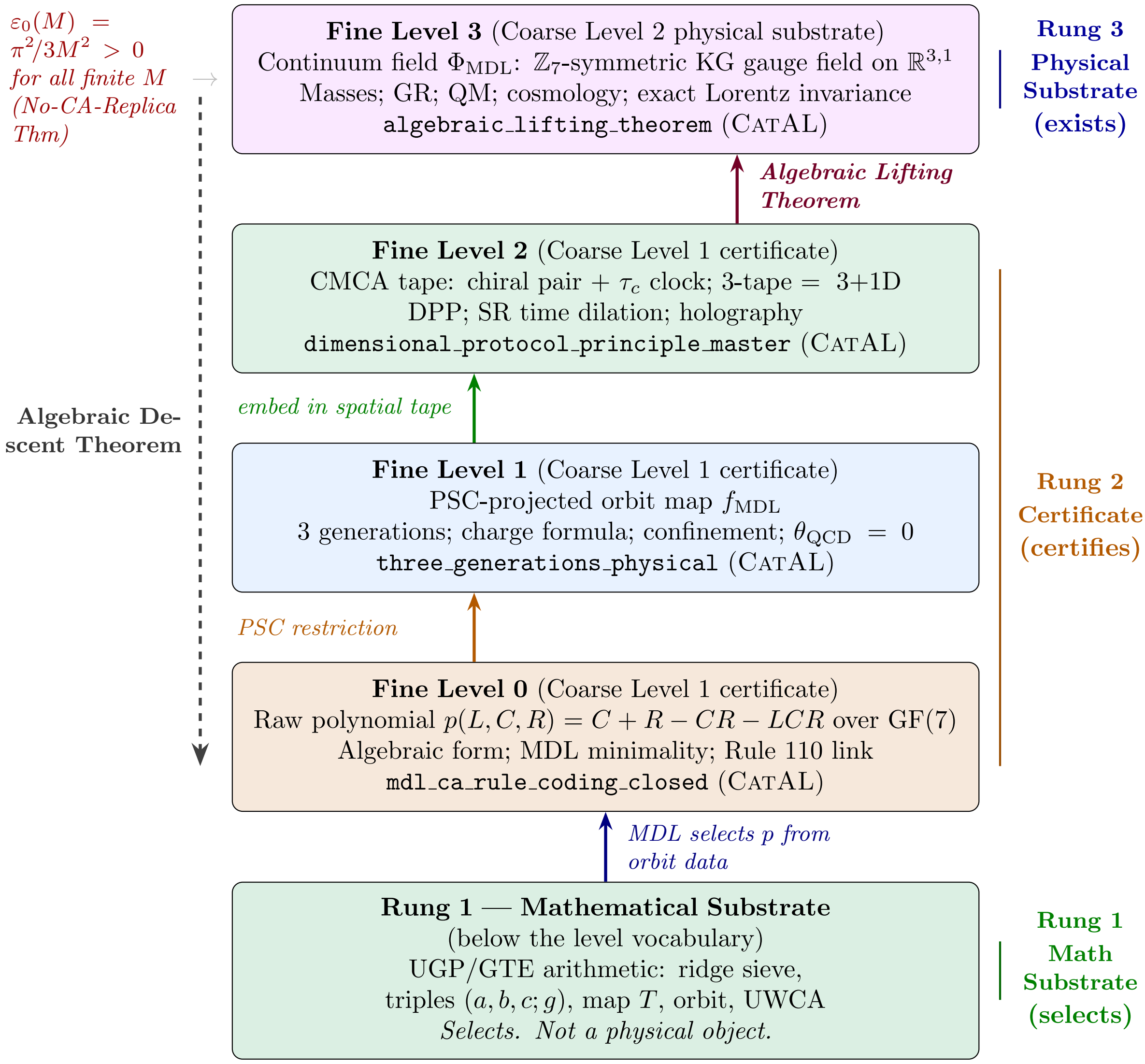
Claim: Einstein's equations $G_{\mu\nu} = 8\pi G\, T_{\mu\nu}$ emerge from the PMDL variational principle applied to Φ_{MDL}.
Level: 3 (continuum field theory — the physical substrate).
Why level 3: This is a statement about the dynamics of the continuum field Φ_{MDL} and its variational principle. It requires the Lifting Theorem to carry the Level-2 algebraic structure to Level 3.
Cert: CatAD (full analytic derivation; computationally confirmed; Lean proof of key sub-steps in progress).

Mixed-level example: particle masses

Claim: The electron mass is $m_e = 0.511\,\mathrm{MeV}$ (from the GTE cascade).
Level: Mixed (Level 1 algebraic structure + Level 3 continuum field).
Why mixed: The *identity* of the electron (which winding sector it belongs to) is a Level 1 fact. The *mass* requires the Level-3 orbit cascade: the energy of a Level 3 kink with winding $w = 4$. The cascade arithmetic produces $N_{\mathrm{eff}} = 73$ (from the Lepton Seed), and the Level 3 kink energy formula converts this to a mass.
Cert: CatAD (0.295% RMS vs PDG 2022 for all nine charged-fermion masses; no free parameters).

9 Summary Diagram: The Full Structure at a Glance

The following TikZ figure shows all three ladder rungs and four fine levels in one picture, with arrows indicating the Lifting and Descent directions.



Reading the diagram:

- The **solid upward arrows** on the left show the derivation chain: the mathematical substrate (Rung 1) produces the polynomial (Level 0) via MDL selection; the polynomial is restricted to PSC-admissible inputs to give the orbit map (Level 1); the orbit map is embedded in a spatial tape (Level 2); and the Algebraic Lifting Theorem lifts the tape's discrete structure to the continuum field (Level 3).
- The **dashed downward arrow** on the right shows the Descent Theorem: the algebraic structure certified at Level 1–2 must be respected by any continuum model at any resolution.
- The **curly braces** on the right label the two ontological rungs. Rung 1 (Mathematical Substrate) is at the bottom; the level vocabulary starts above it. Levels 0–2 are all on Rung 2 (Certificate). Level 3 alone is on Rung 3 (Physical Substrate).
- The **No-CA-Replica** annotation explains why Level 2 cannot be promoted to Level 3 by itself: any finite-resolution CA has a non-zero Lorentz error.

10 Summary

The key things to remember

1. **The coarse two-level scheme** (used in most papers): Level 1 = algebraic certificate (discrete, machine-certifiable); Level 2 = physical substrate (Φ_{MDL}, continuum, exists).

2. **The fine four-level scheme** (used in detailed derivations): Level 0 (raw polynomial) $\to$ Level 1 (f_{MDL} orbit) $\to$ Level 2 (CMCA tape) $\to$ Level 3 (Φ_{MDL} field). In the coarse scheme, Levels 0–2 all collapse to "Level 1 certificate."

3. **The ontological ladder** has three rungs: Mathematical Substrate (selects, below the level vocabulary), Certificate (certifies, = fine Levels 0–2), Physical Substrate (exists, = fine Level 3). Levels grade *resolution*; rungs grade *role*.

4. **The Algebraic Lifting Theorem** (`algebraic_lifting_theorem`, CatAL, zero sorry): every property certified at the discrete Level 1 must exist in the continuum Level 3 field. This is why proving a theorem about the CA is proving a theorem about particle physics.

5. **The Algebraic Descent Theorem** (`algebraic_descent_theorem`, CatAL, zero sorry): every F_{21} conservation law certified at the discrete level holds in any continuum realization at any resolution $M \geq 1$. The algebraic facts are M-independent.

6. **The No-CA-Replica Theorem** (`no_finite_ca_exact_lorentz_replica`, CatAL, zero sorry): no finite-resolution CA achieves exact Lorentz invariance. Only Φ_{MDL} (the $M \to \infty$ continuum limit) achieves $\varepsilon_0 = 0$ exactly. The CA is not the universe; it is the universe's proof engine.

7. **Level framing in papers:** every result carries a level label. Level 0–2 results are in principle machine-verifiable (CatAL). Level 3 results additionally require the Lifting Theorem (CatAD or above).

The full derivation chain is documented across the GTE paper series (P00–P51) and machine-certified in over 300 Lean 4 modules (zero sorry, zero custom axioms), all publicly available at https://zenodo.org/records/20560550 and https://novaspivack.com/research.

Further Reading

Primary Sources

The results in this tutorial are derived and certified in the following papers, all available open-access on Zenodo and at novaspivack.com/research:

- **P48 — The Complete GTE Framework** (main synthesis monograph; Chapter 4 Mathematical Substrate, Chapter 6 One Field — Lifting and Descent, four-level architecture box in Chapter 6):
  doi.org/10.5281/zenodo.20560550
- **P41 — The Three-Layer Chiral Minkowski CA** (Level 2 CMCA architecture):
  doi.org/10.5281/zenodo.20417572
- **P43 — The Complete Φ_{MDL} Framework** (no-CA-replica theorem; Lifting and Descent theorems):
  doi.org/10.5281/zenodo.20417578
- **P45 — The Three-Tape CMCA** (Dimensional Protocol Principle; 3+1D architecture):
  doi.org/10.5281/zenodo.20465805
- **P46 — The GTE Polynomial as UFT** (five simultaneous physical roles; Level framing):
  doi.org/10.5281/zenodo.20465809

Lean 4 machine proofs: github.com/novaspivack/ugp-lean

References

[1] Nova Spivack. The GTE polynomial: A step-by-step cheat sheet: UGP physics tutorial series, 2026. URL `https://doi.org/10.5281/zenodo.20657889`. UGP Physics Tutorial Series.

[2] Nova Spivack. The Φ_{MDL} field: From discrete certificate to continuum physics: UGP physics tutorial series, 2026. URL `https://doi.org/10.5281/zenodo.20657885`. UGP Physics Tutorial Series.

[3] Nova Spivack. From the arithmetic substrate to particle masses: UGP physics tutorial series, 2026. URL `https://doi.org/10.5281/zenodo.20657913`. UGP Physics Tutorial Series.

[4] Nova Spivack. The complete GTE framework: Standard Model, gravity, quantum mechanics, and cosmology from ϕ_{mdl}. Zenodo, 2026. URL `https://doi.org/10.5281/zenodo.20560550`. `https://doi.org/10.5281/zenodo.20560550`. Preprint.